

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

Barcode Area

Front Page Bar Code

29 MAY 2018 (a.m.)

FILL IN ALL THE INFORMATION REQUESTED CLEARLY IN CAPITAL LETTERS.

TEST CODE

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SUBJECT BIOLOGY – UNIT 2 – Paper 02

PROFICIENCY ADVANCED

REGISTRATION NUMBER

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SCHOOL/CENTRE NUMBER

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NAME OF SCHOOL/CENTRE

CANDIDATE'S FULL NAME (FIRST, MIDDLE, LAST)

DATE OF BIRTH

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SIGNATURE _____

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FORM TP 2018156



TEST CODE **02207020**

MAY/JUNE 2018

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

BIOLOGY

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. You may use a silent, non-programmable calculator to answer questions.
5. You are advised to take some time to read through the paper and plan your answers.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) An outline of the Krebs cycle is given in Figure 1, with compounds labelled A to G and selected points in the cycle numbered 1 to 5.

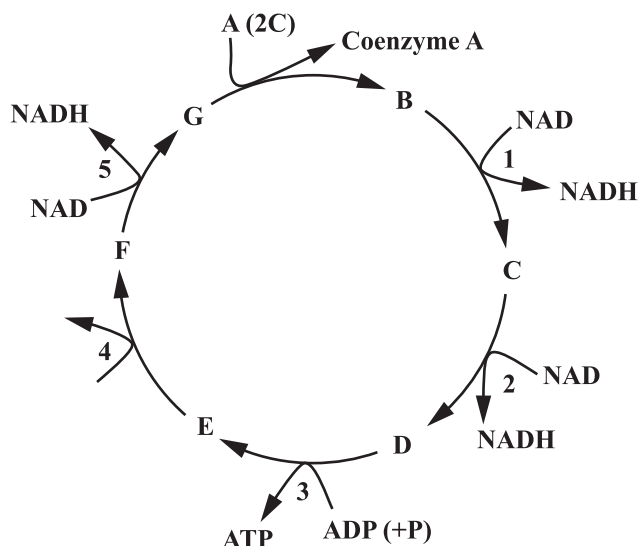


Figure 1. Outline of the Krebs cycle (NADH = reduced NAD)

- (i) **Writing your answer in Figure 1**, insert the number of carbon atoms contained in EACH of the compounds labelled B to G. An example is given for Compound A. **[3 marks]**
- (ii) **Writing your answer in Figure 1**, insert the input and product at the point numbered 4, as completed for the points numbered 1, 2, 3 and 5. **[1 mark]**

GO ON TO THE NEXT PAGE

- (iii) Explain the TWO MAIN roles of the Krebs cycle.

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[4 marks]

- (b) The rate of photosynthesis is measured in individual leaves of a plant species at different light levels, and the data obtained are recorded in Table 1.

**TABLE 1: RATE OF PHOTOSYNTHESIS IN INDIVIDUAL LEAVES
AT DIFFERENT LIGHT LEVELS**

Light Level (Standard Units)	Rate of Photosynthesis (Standard Units)
75	0
150	4
225	8
300	11
400	15
500	18
700	22
900	24
1200	25
1400	25
1600	25

- (i) On the grid provided on **page 7**, plot a line graph based on the data provided in Table 1. **[4 marks]**
- (ii) Using the concept of limiting factors, suggest **THREE** reasons for the shape of the graph drawn in b (i).

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[3 marks]

Total 15 marks

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2. (a) Figure 2A shows a cross-section of phloem tissue and Figure 2B shows a longitudinal section of phloem tissue.

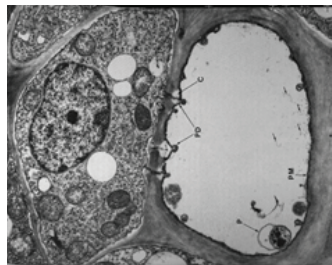


Figure 2A. Cross-section of phloem tissue

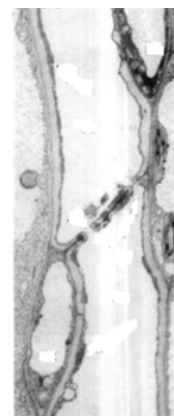
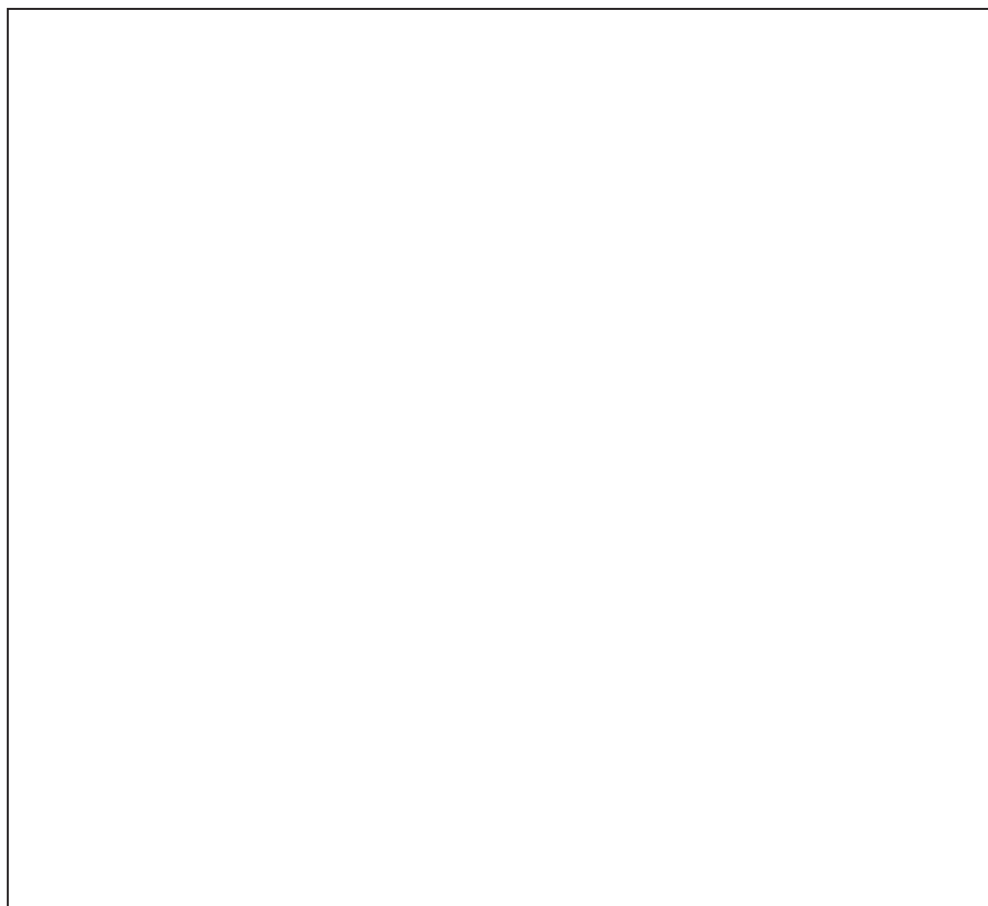


Figure 2B. Longitudinal section of phloem tissue

- (i) In the space below, draw and label a diagram of the TWO phloem cell types shown in Figure 2B.



[3 marks]

GO ON TO THE NEXT PAGE

- (ii) Describe TWO features of EACH phloem cell type shown in Figure 2B.

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[2 marks]

- (iii) State the **major** function of EACH of the TWO phloem cell types.

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[2 marks]

GO ON TO THE NEXT PAGE

- (b) John is a 62-year-old male who recently began using the antidepressant fluoxetine, to deal with the loss of his wife. His blood pressure and pulse rate are monitored for one hour after he ingests the antidepressant. Ten minutes into the monitoring process, John falls asleep. Blood tests show that 30 minutes after ingestion, fluoxetine enters John's bloodstream. Figure 3 shows the data collected on John's systolic and diastolic blood pressure, and his pulse rate over the one-hour period.

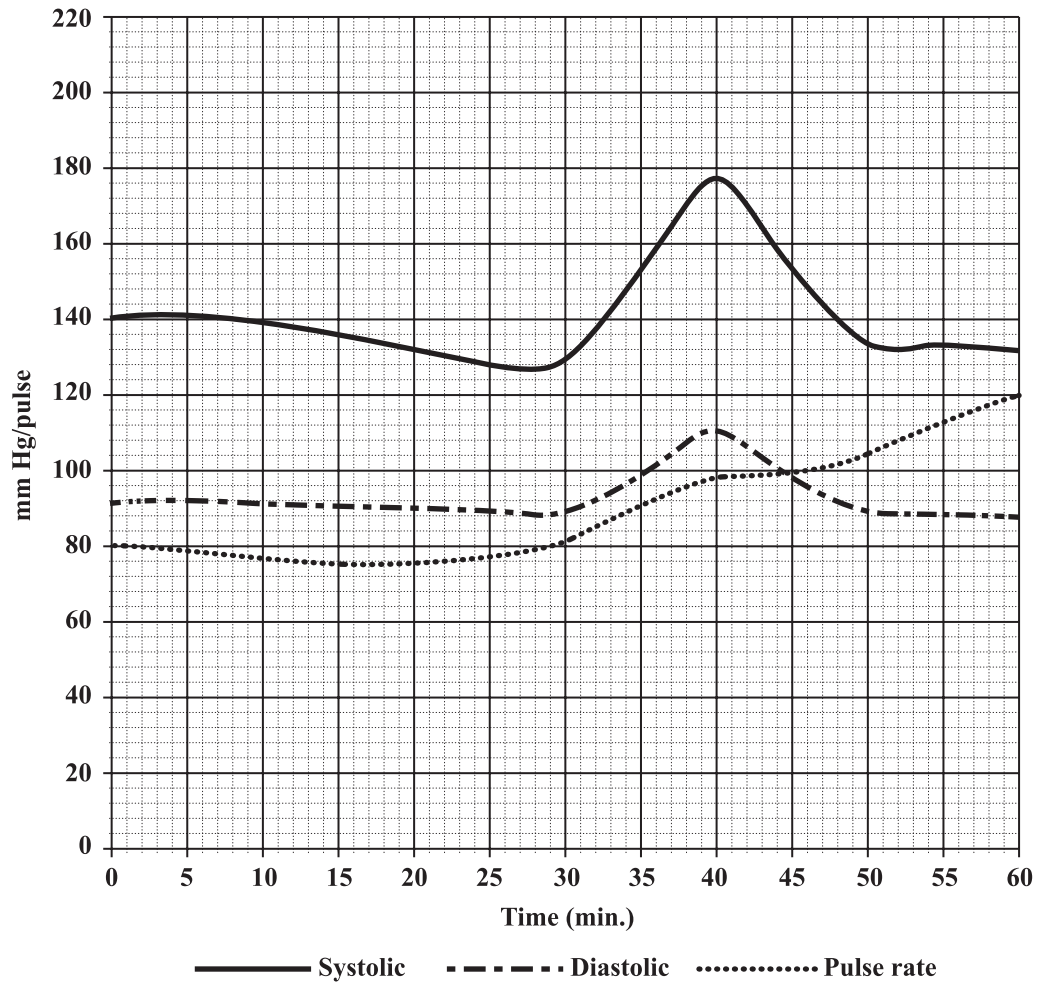


Figure 3. Systolic and diastolic blood pressure and pulse rate over a one-hour period

GO ON TO THE NEXT PAGE

- (i) Explain what is meant by EACH of the following terms:

Blood pressure

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[2 marks]

Pulse

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[2 marks]

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- (ii) Describe the effect of EACH of the following parameters on blood pressure and pulse rate:

Sleep

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[1 mark]

Fluoxetine

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[1 mark]

- (iii) Should John continue to take fluoxetine? Give a reason for your answer.

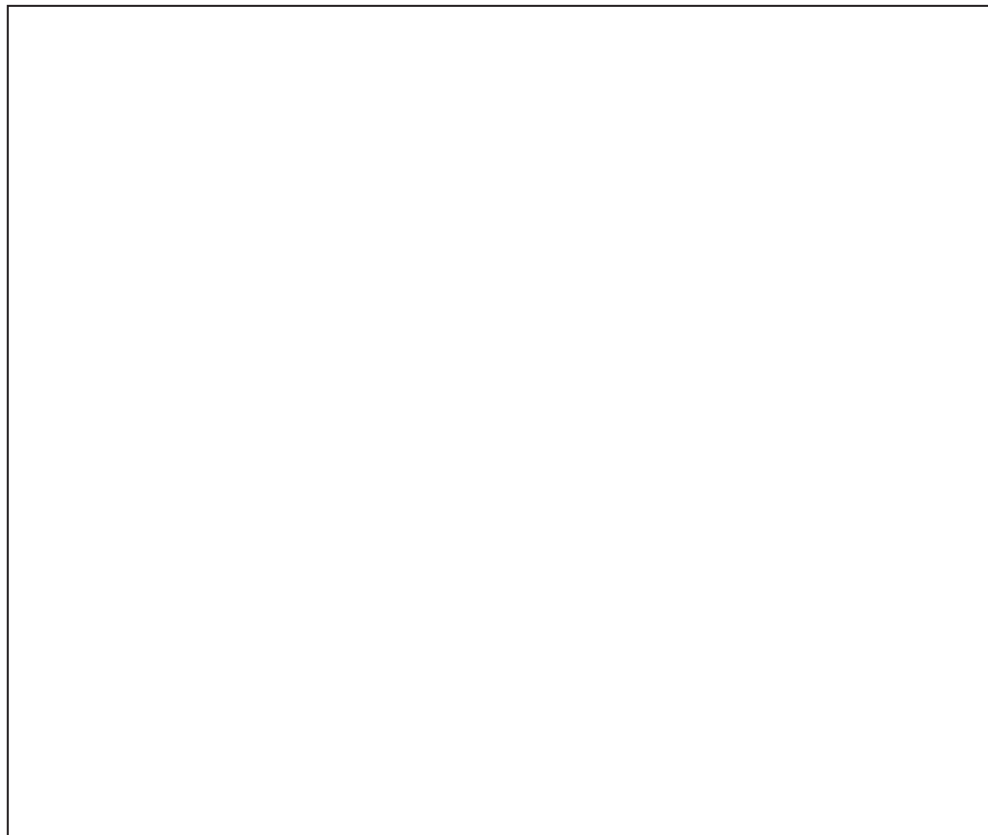
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[2 marks]

Total 15 marks

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3. (a) Alcohol consumption can have short-term and long-term effects on the liver.

- (i) **In the box provided below**, draw a flowchart to illustrate the stages in the detoxification of alcohol in the liver, including the enzymes and coenzyme involved.



[4 marks]

- (ii) Explain how the metabolism of alcohol can lead to a short-term effect on the liver.

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[3 marks]

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- (iii) Describe how alcohol-induced cirrhosis occurs and its effects on the liver.

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[2 marks]

- (b) (i) Complete Table 2 by writing the numbers 2, 3, 4, 5 and 6 to indicate the order of the steps in the replication of HIV in host cells. Steps 1 and 7 have been done for you.

TABLE 2: STEPS IN THE REPLICATION OF HIV IN HOST CELLS

Transmission	1
Transcription	
Reverse transcription	
Maturation and release	7
Integration	
Infection	
Translation	

[3 marks]

GO ON TO THE NEXT PAGE

- (ii) Describe THREE ways by which HIV can be transmitted.

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[3 marks]

Total 15 marks

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Total 15 marks

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Total 15 marks

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CANDIDATE'S RECEIPT

INSTRUCTIONS TO CANDIDATE:

1. Fill in all the information requested clearly in capital letters.

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SUBJECT: BIOLOGY – UNIT 2 – Paper 02

PROFICIENCY: ADVANCED

REGISTRATION NUMBER:

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FULL NAME: _____
(BLOCK LETTERS)

Signature: _____

Date: _____

2. Ensure that this slip is detached by the Supervisor or Invigilator and given to you when you hand in this booklet.
3. Keep it in a safe place until you have received your results.

INSTRUCTION TO SUPERVISOR/INVIGILATOR:

Sign the declaration below, detach this slip and hand it to the candidate as his/her receipt for this booklet collected by you.

I hereby acknowledge receipt of the candidate's booklet for the examination stated above.

Signature: _____
Supervisor/Invigilator

Date: _____